

BREAKHOOK · PROBLEM STATEMENT

BreakHook problem statement

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DOCUMENT CONTROL

Document ID	BHK-PRB-001	Version	0.2
Status	Draft	Date	2026-10-03
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/breakhook		
Source	docs/01-problem.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-03	Amish Chadha	TRL 2 and TRL 3; co-design candidate named; open questions settled under Amish's 2026-10-03 pre-approval (BHK-DDR-001); safety section added
0.1	2026-09-30	Amish Chadha	Initial scaffold

BreakHook problem statement

In the first minutes of a settlement fire, only residents are there. They know that clearing a gap can stop the spread, but they have no tool that lets them do it from a safe distance.

■ The problem

Fire can spread between informal dwellings in under a minute ([Stellenbosch University, 2023](#)). Fire engines often cannot get in: response times average 68 minutes in informal settlements in the Dhaka and Cape Town study, and pathways can be 60 to 90 cm wide ([Royal Academy of Engineering](#)). By then a fire can take hundreds or thousands of homes, as at Imizamo Yethu ([Imizamo Yethu fire spread analysis](#)).

Current responses focus on prevention and access: fire safety education, extinguishers and re-blocking into rows with 3 m gaps, which reduces settlement capacity by at least 20 % ([The New Humanitarian, 2013](#)). Community responders, trained by groups such as the Kenya Red Cross, use buckets and basins and pull structures down by hand ([Ngau and Boit, 2020](#)). The historical fire hook was built for exactly this job ([London Fire Brigade Museum](#)), but no open, tested version exists for these communities.

■ Users and context

User	Need	Context
Community fire responders	Pull a structure down from a safe distance to open a firebreak	First minutes of a fire, crowds, smoke, narrow paths
Settlement committees and ward leaders	A tool they can store, assign and govern, with a clear rule on when it may be used	Community fire plans and training days
Municipal fire services and reservist programmes	Equipment to issue to trained community reservists	City fire and disaster management
NGOs and Red Cross societies	A low-cost kit item with training material	Community fire response training

■ Operating environment

- Dense settlements of closely spaced, mostly single-storey dwellings (assumed; to be confirmed with the partner).
- Paths about 0.6 to 0.9 m (2 to 3 ft) wide in places ([Royal Academy of Engineering](#)).
- Smoke, radiant heat, crowds and panic during use.
- Overhead and informal electrical wiring may be present.
- Stored outdoors or in a shared shed for long periods between uses.

■ Constraints

- Prototype parts budget: USD 2,000 or less for a community kit.
- Pole must be non-conductive along the handled length.
- Light enough to carry and for a team of two to lift into place: 8 kg (18 lb) or less for the pole and hook head.

- Short enough in sections to move along narrow paths.
- Buildable by a local welder from stock steel plus commercial fibreglass tube and rope.
- Open design: hardware under CERN-OHL-S-2.0, training material under a matching open licence.

■ Out of scope

- Fire suppression equipment, alarms and detectors.
- Demolition outside an active fire emergency.
- Multi-storey or masonry buildings.
- Rules on who may authorise pulling down a home; that is for each community and its local authority.

■ Prior work

Prior work	What it does	Gap for these users	Source
Historical fire hooks	Long hooks used to pull down buildings in the path of a fire, as in London in 1666	No modern, documented, non-conductive version sized for informal dwellings	link
Community fire response in Nairobi	Residents trained by the Kenya Red Cross use buckets and tear down structures to make firebreaks	Done by hand, close to the fire, with no purpose-made tool	link
Re-blocking (Cape Town)	Rebuilding settlements in rows with 3 m gaps for access and fire separation	Slow, costly, reduces capacity; no help once a fire has started	link

■ Co-design

A municipal fire service reservist programme or a Red Cross or Red Crescent society already training community fire responders in informal settlements, working with a settlement committee that can set rules for when the tool is used.

First candidate to approach (not agreed): the City of Cape Town Fire and Rescue Service and its volunteer reservists, with a settlement committee in the Western Cape. Second candidate: the Kenya Red Cross Society's community fire response work in Nairobi (BHK-DDR-001).

■ Questions from TRL 1 and where they were settled

All five were decided on 2026-10-03 under Amish's pre-approval and are recorded in the design decisions register (BHK-DEC-001) and decision record BHK-DDR-001.

Question	Decision
What stops misuse, for example in disputes or evictions?	The kit is held by the settlement committee in a sealed, locked rack with two named keyholders; every use is logged against the seal number; a pull happens only during a fire, inside a fire plan agreed with the fire service, on a caller's word
How far ahead of the fire must the break be?	Training rule: open the break at least two dwellings ahead of the burning one, and only if the pull can finish before the fire reaches it; otherwise evacuate and wait for the fire service
A pulley or capstan to cut the crew?	Not in the base kit. A pulley needs a strong anchor that is rarely there; the kit uses ten hauling toggles instead
Shared walls?	The tool is used only on free-standing dwellings or the end unit of a row; rows with shared walls are outside its scope
Who stores and checks the kit?	The settlement committee, on a monthly check card: seal intact, pins and toggles present, no cracks in the tube, rope dry and unfrayed; a broken seal means a full check and a fresh proof load before reuse

■ Safety

Safety: BreakHook is used to pull down a structure with a hook, a pole and a rope, near a fire. The hazards are a structure falling on people, a dwelling that is not empty, overhead and informal electrical wiring touching the metal hook or a wet rope, a rope or fitting breaking under load, heat and smoke, and crowds. The design answers with a pole that is withdrawn before anyone hauls, haulers at least 10 m away, a rated rope system proof-loaded before issue, an insulated pole length used only well away from wires, and a check card that must be completed before every pull. It is published as an open engineering reference, not certified equipment, and is used only within a community fire plan agreed with the local fire service.